

MATH 351 Project report

“Using Lagrange Interpolating polynomial and Natural Cubic Spline to predict the population of Kazakhstan and observe valuable insights from the data”

Meirzhan Kurmanov
Aigerim Tursynbekkyzy
Maxim Kabakin

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Constructing a Lagrange polynomial and Natural Cubic Spline polynomials.

Initially, we were provided with the following dataset on the population of Kazakhstan between 2019 and 2024.

Year	2019	2020	2021	2022	2023	2024
All population	18,395,567	18,631,779	18,879,159	19,503,159	19,766,807	20,033,842
Urban population	10,698,208	10,938,652	11,151,376	11,991,238	12,209,896	12,451,192
Rural population	7,697,359	7,693,127	7,728,176	7,511,921	7,556,911	7,582,650

Table 1. Initial data on population of Kazakhstan

In both interpolation models, the data from 2020 to 2023 was utilized to predict the number of the population for the year 2024. The overall number of datapoints is therefore 4.

Lagrange polynomial results - predicting population for 2024

	Predicted population (2024)	Actual population (2024)	Abs. error	Rel. error
All population	18,974,703	20,033,842	1,059,139	0.05287
Urban population	10,559,008	12,451,192	1,892,184	0.15200
Rural population	8,375,695	7,582,650	793,045	0.10460

Table 2. Predicted data for 2024 by Lagrange interpolating polynomial

As it's observed, the predicted results didn't quite match the actual population, yielding a relative error of roughly 5%, 15% and 10% for the entire, urban and rural population respectively.

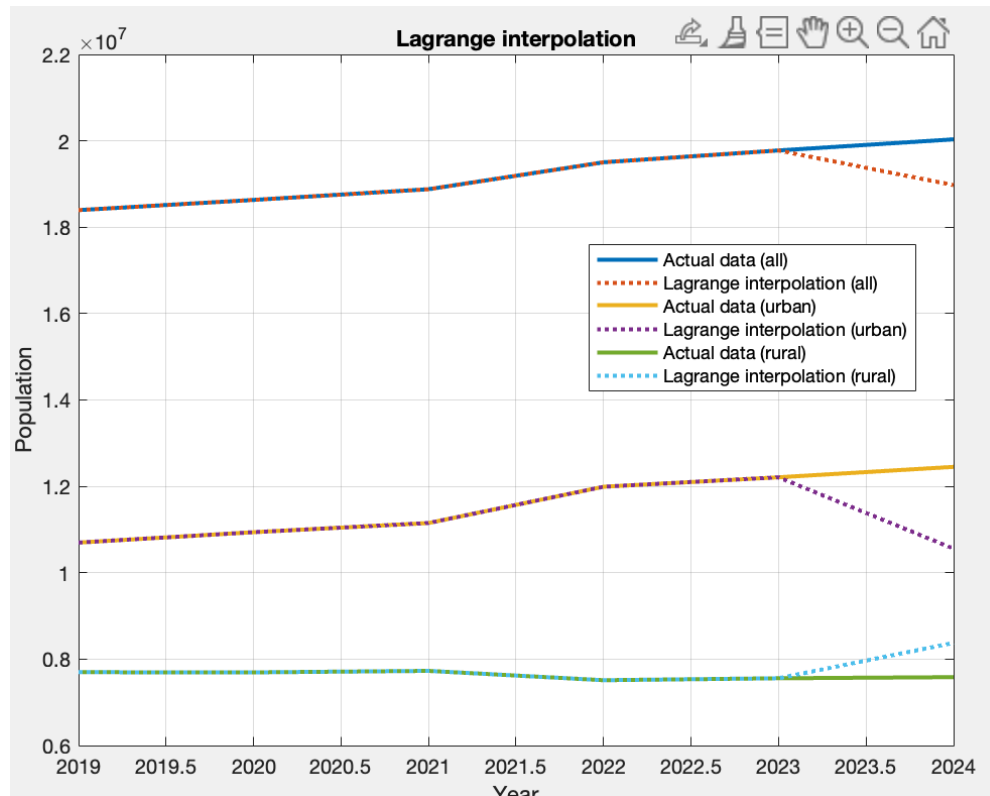


Figure 1. Comparison of the Lagrange interpolation results to the actual data

Natural Cubic Spline results - predicting population for 2024

	Predicted population	Actual population	Abs. error	Rel. error
All population	20,050,455	20,033,842	16,613	0.00083
Urban population	12,428,554	12,451,192	22,638	0.00182
Rural population	7,601,901	7,582,650	19,251	0.00254

Table 3. Predicted data for 2024 by Natural Cubic Spline

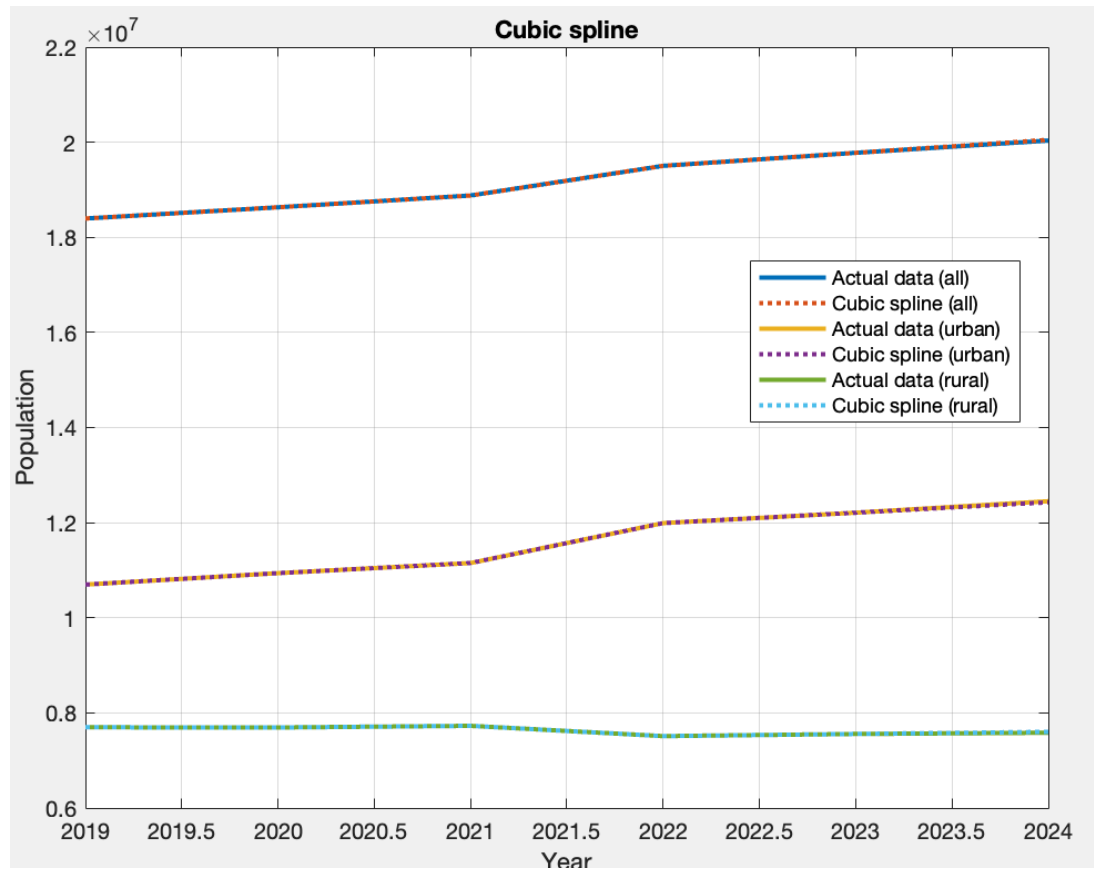


Figure 2. Graphical comparison of the Natural Cubic Spline results and actual data

Data analysis

Natural Cubic Spline yielded very small errors compared to the Lagrange interpolating polynomial, having relative errors of less than 0.3% for each type of the population, while relative error for Lagrange interpolation varied from 5% to 15%.

The possible reason for such relatively poor performance of the Lagrange interpolation on predicting the population in 2024 could be the fact that this method may show strong fluctuations when we are trying to approximate points that go beyond the range of the initial data (from 2020 to 2023), mainly because Lagrange polynomial is a single polynomial of higher degree and the more points we add, the more likely fluctuations are to occur when we're stepping out of the initial range. Cubic spline, on the other hand, is a method that is supposed to handle such difficulty. Because of its piecewise nature, it allows the construction of distinct cubic polynomials for each subinterval of data points meaning that fluctuations are much less likely to happen.

However, as it's observed from the Figure 1, Lagrange interpolation handled the intermediate values, that is, the points between $x = \{2020, 2021, 2022, 2023\}$ quite well, because the Weierstrass Approximation theorem suggests that for any function, say $f(x)$ (a function that describes the population of the Kazakhstan) there always exists an approximating polynomial $g(x)$ on the interval $[a,b]$ such that for all $\epsilon > 0$, $|f(x) - g(x)| < \epsilon$ for all x in $[a,b]$, in our case, for all x in $[2020, 2023]$. Indeed, such a result is observed on Figure 1.

To predict the number of population for $x = 2024$, which falls beyond the initial range, the Cubic Spline is a preferred option due to the lack of fluctuations and much smaller error in approximation. For example, the mean of the relative error of the cubic spline is almost 60 times smaller than that of the Lagrange polynomial.

When will the population reach 20.2 mln?

Based on the data from the cubic Spline interpolation and by utilizing the data points between 2020 and 2024, the population is expected to hit the mark of 20.2 mln in the 3-rd month (March) in 2025.

It should be noted that if we utilize the data from 2019 to 2024, then we would reach this population in 7-th month (June) of 2024.

Population prediction for 2026

In the data analysis part, it was already mentioned why Cubic spline would be a much preferable option to estimate the number of population that goes beyond the initial range (2019 - 2024)

If we would utilize all data from 2019 to 2024, then the predicted urban population for 2026 would be 12,557,074. For the same model, the predicted rural population will be 7,807,560. However, if we shorten the range of the initial data, and utilize values only between 2020 and 2023 as in our initial model, then the urban population is projected to be 10,452,789 and the rural population will be 8,534,351.

So the predicted population for 2026, according to cubic spline results, strongly depends on the range of the initial values we decide to pick. It's clearly seen that for data between 2020 and 2023, the forecast is much more optimistic towards rural population - it increased by 12.5% since 2024 - while utilization of data between 2019 and 2024 shows that the general trend for the slower growth of rural areas is kept, showing only 3% growth in 2 years.

The ways to solve this problem will be provided later in the report.

Which model is better?

We constructed 2 models to predict population for 2024 using Lagrange interpolation polynomial and cubic spline interpolation. From the results above in the data analysis part, we can witness that Model 2 has less absolute error compared to Lagrange interpolation [Table 2 and 3], therefore has a higher accuracy. Graphical comparisons of model 1 and model 2 shown above [Figure1, 2] are clear demonstrations of the accuracy of methods, where from Figure1 it can be observed that the data from the cubic Spline interpolation is much closer to the actual model. Not to mention, cubic spline uses a piecewise cubic polynomial and provides a smooth approximation of the data. At this point, Dyer S. and Dyer J. refer to the cubic spline as a very effective, well-behaved, computationally efficient interpolator [1]. Therefore, in terms of accuracy cubic spline interpolation is a better approach for our study.

Alternative methods

Besides the polynomial interpolation methods, we would like to present two most interesting ones among numerous methods that can be implemented in predicting population.

1. Time series Forecasting

Time series forecasting predicts values in an unknown future based on a series of predictors varying over time. This method has potential in forecasting in numerous areas like economics, sales and even budgetary analysis that requires high accuracy. J Brownlee in his book “Introduction to time series forecasting with python” describes it as an important area of machine learning that has a huge advantage in predicting population [2]. Time series forecasting methods explicitly take into account the time-ordered nature of the data, which is crucial for population prediction. Furthermore, this method was used in predicting the population in Kenya by Samuel Kimani from The University of Nairobi, where he derived quite accurate and important results [3].

2. The Exponential Smoothing Method

Another method that we can apply for population prediction is the exponential smoothing. It is a linear forecasting method, which was first proposed by American scholar R.G.Brown in 1959. This method has a clear process and is easier to implement as it requires less parameters compared to time series forecasting. The Exponential Smoothing Method is widely used in economics and management forecasting to predict the future data from the history data. In the work of Yi Huang [4], where he applied The Exponential Smoothing method in forecasting GDP (Gross Domestic Product) of China he explained the properties of this method. He mentioned three types of this method, however the primary (simple) exponential smoothing method (SES) , that does not include trend and seasonality, is mostly used and also suitable for our study. Yi

Huang recommends combining multiple methods to improve accuracy of forecasting like every other method that can be used for forecasting [4].

The formula for simple exponential smoothing is:

$$S_{(t)} = \alpha x_{(t)} + (1 - \alpha)S_{(t-1)}$$

- $s_{(t)}$ is the smoothed value at time t ,
- $x_{(t)}$ is the observed value at time t ,
- s_{t-1} is the previous smoothed statistic, and
- α is the smoothing parameter between 0 and 1.

According to Professor Thomas B. Fomby from Southern Methodist University, pretty accurate forecasts are likely via exponential smoothing method as well-chosen type of this method will be less time consuming compared to other methods [5]. Another advantage he mentions is that the exponential smoothing does not require a long history of data to predict future values.

To conclude, for general population prediction, especially if simplicity and clear process are desired, which is our case, Exponential Smoothing is a reliable choice. When it comes to more detailed analysis that needs to take into account complex parameters for long-term forecasts, time series models are often the better choice. Therefore, for population prediction in our study the Exponential Smoothing will be a more preferable method.

The ways to solve the problem

1. Rural areas

As it has already been discussed, in order to project the rural population in Kazakhstan for the year 2026 we could use the cubic spline interpolation as the model, and obtain an absolute error of 726,791 and a relative error of 9.31% - the errors obtained on the differences of ranges we would choose, as it has been mentioned.

Nonetheless, according to our result that we have got, the rural population in Kazakhstan would experience moderate growth in 2026 (by 205,659), and the reason why that would have been the case is that this is actually a global tendency - even though, the birth rates in villages are relatively higher [6], people from rural areas are more prone to migrate towards cities [7], hence even more people result becoming the part of an urban population. As for Kazakhstan, we could name some other causes of such a situation, such as a poorer infrastructure, the fact of worse access to worthy healthcare, and much more limited economic opportunities in the rural areas [8].

Therefore, had we been investors, we would have been obliged to treat these problems by developing the infrastructure, proposing some technological enhancement in agriculture and industry, in order to provide more people with workplaces, and building modern venues of healthcare, wherever possible [9]. Besides, we could come up with some more creative solutions, such as encouraging the development of small-scale agricultural businesses, popularizing the idea of rural tourism (just like in neighboring Kyrgyzstan) [10] and even investing in renewable energy [11]. These suggestions would contribute to a more successful future of rural areas and a better standard of living for the rural population.

2. Urban areas

As it has already been mentioned before, the urban population in Kazakhstan in 2026 would probably be equal to 12,557,074 - a rather impetuous growth of the urban population by that year, and that would certainly make sense.

It can also be considered a global trend of the rapid growth of population in urban areas, and we could say that it is the consequence of the conclusions we made in the previous question [12]. Thus, if the healthcare system or infrastructure tend to deteriorate in rural neighborhoods [13], the same components, on the contrary, seem to prosper in the urban areas [14], and surely people appear to be attracted to these parts of their countries. That is particularly seen in Kazakhstan, where the difference in the quality of life between the major cities, like Almaty or Astana, and our villages speaks for itself [15].

Hence, we, as investors, would only help our cities develop even more, so that their sustainability would be justified in the long term [16]. That would mean that we would have invested in the features of a future urban area, such as urban digital technologies, together with some technological startups [17], energy-saving venues and, probably, more ecological public transportation [18] - this would all contribute to the diversification of economic sectors in urban areas [19], as well as to the well-being of urban populations.

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Appendix: Code

```
%-----Task 1-----
function y_pred = lagrange_interpolation(x, x_points, y_points)
    y = 0;
    n = length(x_points);
    for i = 1:n
        p = 1;
        for j = 1:n
            if i ~= j
                p = p * (x - x_points(j))/(x_points(i) - x_points(j)); %Product
notation formula
            end
        end
        y = y + p * y_points(i);
    end
    y_pred = y;
end
function y_pred = cubic_spline (x, x_points, y_points)
    a_values = y_points;
    n = length(x_points) - 1; % number of subintervals
    %Step 1
    h_values = zeros(1, n);
    for i = 1:n
        h_values(i) = x_points(i+1) - x_points(i);
    end
    %Step 2
    alpha_values = zeros(1, n+1);
    for i = 2:n
        alpha_values(i) = 3 * (a_values(i+1) - a_values(i))/h_values(i) - 3 *
(a_values(i) - a_values(i-1))/h_values(i-1);
    end
    %Step 3
    l_values = ones(1, n+1);
    m_values = zeros(1, n);
    z_values = zeros(1, n+1);
    %Step 4
    for i = 2:n
        l_values(i) = 2 * (x_points(i+1) - x_points(i-1)) - h_values(i-1) *
m_values(i-1);
        m_values(i) = h_values(i) / l_values(i);
        z_values(i) = (alpha_values(i) -
h_values(i-1)*z_values(i-1))/(l_values(i));
    end
    %Step 5
    c_values = zeros(1, n+1);
    %Step 6
    b_values = zeros(1, n);
```

```

d_values = zeros(1, n);
for j = n:-1:1
    c_values(j) = z_values(j) - m_values(j) * c_values(j+1);
    b_values(j) = (a_values(j+1) - a_values(j))/(h_values(j)) - h_values(j)
* (c_values(j+1) + 2 * c_values(j))/3;
    d_values(j) = (c_values(j+1) - c_values(j))/(3 * h_values(j));
end
n = length(x_points); %Now set n to be the number of the points
index = 0;
for i = 2:n
    if x >= x_points(i-1) && x <= x_points(i)
        index = i-1;
        break;
    elseif x > x_points(n)
        index = n-1;
        break;
    elseif x < x_points(1)
        index = 1;
        break;
    end
end
y_pred = a_values(index) + b_values(index) * (x - x_points(index)) +
c_values(index) * (x - x_points(index))^2 + d_values(index) * (x -
x_points(index))^3;
end
% Inputs
year_values = [2019, 2020, 2021, 2022, 2023, 2024];
all_pop = [18395567, 18631779, 18879552, 19503159, 19776807, 20033842];
urban_pop = [10698208, 10938652, 11151376, 11991238, 12209896, 12451192];
rural_pop = [7697359, 7693127, 7728176, 7511921, 7556911, 7582650];
% Task 1
%Lagrange interpolation
disp("Lagrange interpolations results: ");
lpop_all = lagrange_interpolation(2024, year_values(2:5), all_pop(2:5));
lpop_all_abs = abs(all_pop(6) - lpop_all);
lpop_all_rel = round((lpop_all_abs/all_pop(6)), 5);
lpop_urban = lagrange_interpolation(2024, year_values(2:5), urban_pop(2:5));
lpop_urban_abs = abs(urban_pop(6) - lpop_urban);
lpop_urban_rel = round((lpop_urban_abs/urban_pop(6)), 5);
lpop_rural = lagrange_interpolation(2024, year_values(2:5), rural_pop(2:5));
lpop_rural_abs = abs(rural_pop(6) - lpop_rural);
lpop_rural_rel = round((lpop_rural_abs/rural_pop(6)), 5);
formattedStr = sprintf("Predicted population for year %d is %d (Abs. error: %i,
Rel. error: %.3d)", 2024, lpop_all, lpop_all_abs, lpop_all_rel);
disp(formattedStr);
formattedStr = sprintf("Predicted urban population for year %d is %d (Abs.
error: %i, Rel. error: %.3d)", 2024, lpop_urban, lpop_urban_abs,
lpop_urban_rel);
disp(formattedStr);

```

```

formattedStr = sprintf("Predicted rural population for year %d is %d (Abs.
error: %i, Rel. error: %.3d)", 2024, lpop_rural, lpop_rural_abs,
lpop_rural_rel);
disp(formattedStr);
%For cubic spline interpolation
disp("Cubic spline interpolation results:");
spop_all = cubic_spline(2024, year_values(2:5), all_pop(2:5));
spop_all_abs = abs(all_pop(6) - spop_all);
spop_all_rel = round((spop_all_abs/all_pop(6)), 5);
spop_urban = cubic_spline(2024, year_values(2:5), urban_pop(2:5));
spop_urban_abs = abs(urban_pop(6) - spop_urban);
spop_urban_rel = round((spop_urban_abs/urban_pop(6)), 5);
spop_rural = cubic_spline(2024, year_values(2:5), rural_pop(2:5));
spop_rural_abs = abs(rural_pop(6) - spop_rural);
spop_rural_rel = round((spop_rural_abs/rural_pop(6)), 5);
formattedStr = sprintf("Predicted population for year %d is %d (Abs. error: %i,
Rel. error: %.3d)", 2024, spop_all, spop_all_abs, spop_all_rel);
disp(formattedStr);
formattedStr = sprintf("Predicted urban population for year %d is %d (Abs.
error: %i, Rel. error: %.3d)", 2024, spop_urban, spop_urban_abs,
spop_urban_rel);
disp(formattedStr);
formattedStr = sprintf("Predicted rural population for year %d is %d (Abs.
error: %i, Rel. error: %.3d)", 2024, spop_rural, spop_rural_abs,
spop_rural_rel);
disp(formattedStr);
%Plotting for lagrange interpolation
all_modified = all_pop;
all_modified(6) = lpop_all;
urban_modified = urban_pop;
urban_modified(6) = lpop_urban;
rural_modified = rural_pop;
rural_modified(6) = lpop_rural;
figure
plot(year_values, all_pop, '-', 'LineWidth', 2)
hold on
plot(year_values, all_modified, ':', 'LineWidth', 2)
plot(year_values, urban_pop, '-', 'LineWidth', 2)
plot(year_values, urban_modified, ':', 'LineWidth', 2)
plot(year_values, rural_pop, '-', 'LineWidth', 2)
plot(year_values, rural_modified, ':', 'LineWidth', 2)
xlabel('Year')
ylabel('Population')
title('Lagrange interpolation')
legend('Actual data (all)', 'Lagrange interpolation (all)', 'Actual data
(urban)', 'Lagrange interpolation (urban)', 'Actual data (rural)', 'Lagrange
interpolation (rural)')
grid on
hold off

```

```

%Plotting for cubic spline
all_modified = all_pop;
all_modified(6) = spop_all;
urban_modified = urban_pop;
urban_modified(6) = spop_urban;
rural_modified = rural_pop;
rural_modified(6) = spop_rural;
figure
plot(year_values, all_pop, '-', 'LineWidth', 2)
hold on
plot(year_values, all_modified, ':', 'LineWidth', 2)
plot(year_values, urban_pop, '-', 'LineWidth', 2)
plot(year_values, urban_modified, ':', 'LineWidth', 2)
plot(year_values, rural_pop, '-', 'LineWidth', 2)
plot(year_values, rural_modified, ':', 'LineWidth', 2)
xlabel('Year')
ylabel('Population')
title('Cubic spline')
legend('Actual data (all)', 'Cubic spline (all)', 'Actual data (urban)', 'Cubic
spline (urban)', 'Actual data (rural)', 'Cubic spline (rural)')
grid on
hold off
disp(" ")
% ----- Task 4 -----
M = 20200000;
acc = 10;
x = 2019;
h = 0.00001;
while(abs(M - cubic_spline(x, year_values(2:6), all_pop)) > 10)
    x = x + h;
end
target = round(cubic_spline(x, year_values(2:6), all_pop));
year = round(x, 2);
format longG
month = round((abs(year - fix(year)) * 12));
formattedStr = sprintf("The population will reach %i in %i, %i-th month (Cubic
spline)", M, fix(year), month);
disp(formattedStr);
disp(" ")
% -----Task 6 and 7-----
year = 2026;
pred_2026_rural = cubic_spline(year, year_values(1:6), rural_pop);
formattedStr = sprintf("Predicted rural population for year %i, is %.0f (Cubic
spline) - utilizing data from 2019 to 2024", year, pred_2026_rural);
format LongG
disp(formattedStr);
pred_2026_urban = cubic_spline(year, year_values(1:6), urban_pop);
formattedStr = sprintf("Predicted population for year %i, is %.0f (Cubic
spline) - utilizing data from 2019 to 2024", year, pred_2026_urban);

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```
format LongG
disp(formattedStr);
year = 2026;
pred_2026_rural = cubic_spline(year, year_values(2:5), rural_pop);
formattedStr = sprintf("Predicted rural population for year %i, is %.0f (Cubic
spline) - utilizing data from 2020 to 2023", year, pred_2026_rural);
format LongG
disp(formattedStr);
pred_2026_urban = cubic_spline(year, year_values(2:5), urban_pop);
formattedStr = sprintf("Predicted population for year %i, is %.0f (Cubic
spline) - utilizing data from 2020 to 2023", year, pred_2026_urban);
format LongG
disp(formattedStr);
```